13.10.2026

C.W

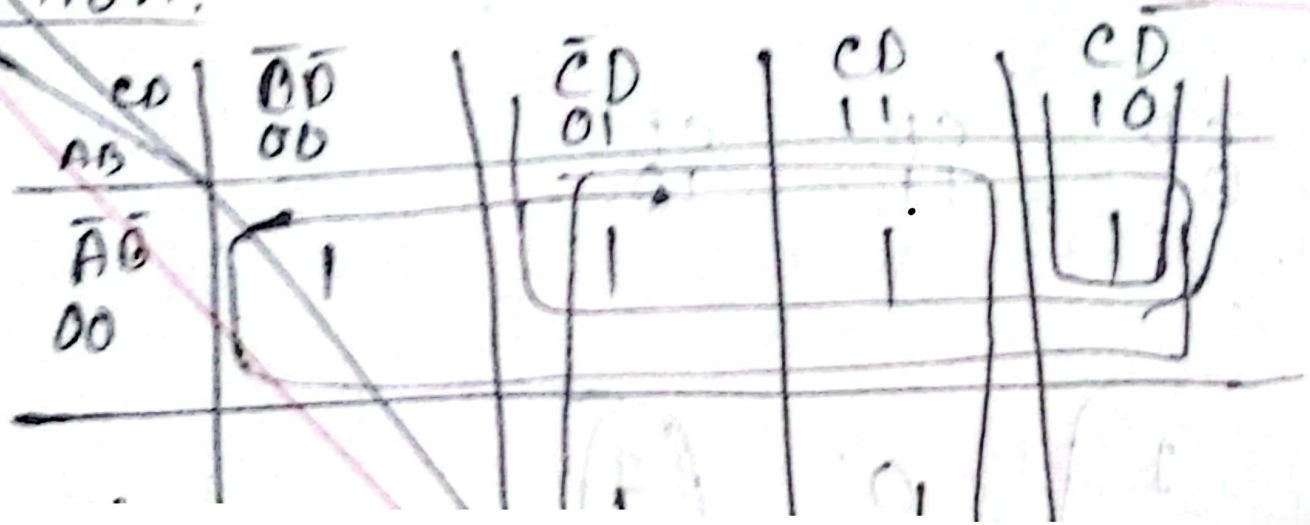
(*) Solve using K-map.

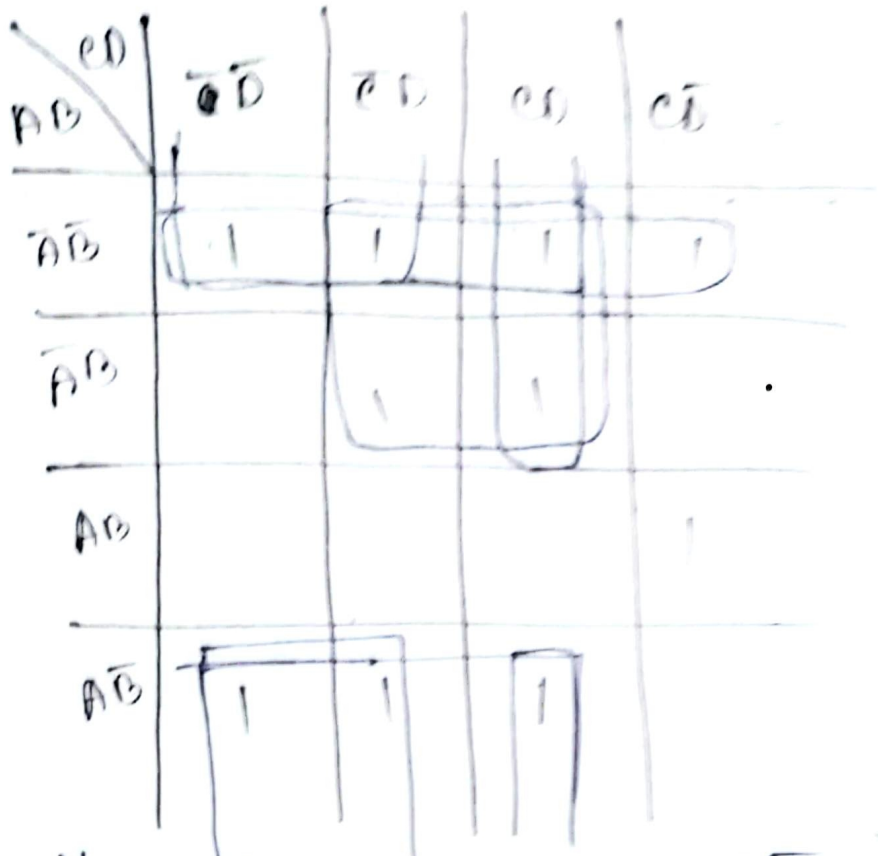
$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD +$$

$$\bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BCD + AB\bar{C}\bar{D} +$$

$$A\bar{B}\bar{C}D + A\bar{B}CD + AB\bar{C}\bar{D}$$

Solution:





$$y = \bar{B}\bar{C}D + A\bar{B} + \bar{A}D + AB\bar{C}\bar{D}$$

$$y = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}D + \cancel{AB\bar{C}\bar{D}}$$

(*) Minimize using K-map:

f) $f(A,B,C,D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 12)$

	A	B	C	D	$\sum m$
0	0	0	0	0	
1	0	0	0	1	
2	0	0	1	0	
3	0	0	1	1	
4	0	1	0	0	
5	0	1	0	1	
6	0	1	1	0	
7	0	1	1	1	

	A	B	C	D	$\sum m$
$A\bar{B}\bar{C}\bar{D}$	1	0	0	0	
$A\bar{B}\bar{C}D$	1	0	0	1	
	1	0	1	0	
	1	0	1	1	
	1	1	0	0	
	1	1	0	1	
	1	1	1	0	
	1	1	1	1	

8
9
10
11
12
14
14
15

HW:

Minimize using K-map:

$f(A, B, C, D) = \Sigma m(0, 2, 4, 8, 12, 13, 15)$

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So, S.D $(\sigma) = \sqrt{\frac{1}{12}(n^2-1)}$

(Showed)

(*) Find IQR for the given data:

Day	0-4	4-8	8-12	12-16	16-20	$n=50$
absent	5	12	23	8	2	
c.f.	5	17	40	48	50	

Solution:

$\therefore Q_1 = L + \frac{\frac{n}{4} - pcf}{f} \times c$

$= 4 + \frac{12.5 - 5}{12} \times 4$

$= 4 + \frac{7.5}{12} \times 4$

$= 4 + 2.5$

$= 6.5$

Here,
$\frac{n}{4} = \frac{50}{4} = 12.5$
$L = 4$
$pcf = 5$
$f = 12$
$c = 4$

$\therefore Q_3 = L + \frac{\frac{3n}{4} - pcf}{f} \times c$

$= 8 + \frac{37.5 - 17}{23} \times 4$

$= 8 + \frac{20.5}{23} \times 4$

$= 8 + 3.5 = 11.5$

Here,
$\frac{3n}{4} = \frac{3 \cdot 50}{4} = 37.5$
$L = 8$
$pcf = 17$
$f = 23$
$c = 4$

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